ChatGPT

Juan Garassino – Comprehensive Biography

Early Life and Education

Juan Garassino (PERSON) was born and raised in Buenos Aires (PLACE), Argentina – a cosmopolitan city known as a UNESCO "Creative City of Design" that blends European architectural heritage with innovative design culture. Growing up in this rich environment, he developed an early passion for architecture and technology.

For his higher education, Juan Garassino STUDIED_AT the **Universidad de Buenos Aires (ORGANIZATION)** Faculty of Architecture, Design and Urbanism (FADU) (TIME_PERIOD: 2012–2018), one of Latin America's leading architecture schools influenced by the French Beaux-Arts tradition. He GRADUATED_FROM FADU with a degree in Architecture (EDUCATION), having specialized in computational design and urbanism (SPECIALIZED_IN). During his studies, he also spent an academic year at the **École Nationale Supérieure d'Architecture Paris-Malaquais (ORGANIZATION)** in Paris (TIME_PERIOD: 2016–2017), an experience that broadened his perspective with European architectural traditions and the French language. Alongside his native Spanish, he became fluent in English during this time, a SKILL he would later use throughout his international career.

Early Career in Architecture and Urbanism

After university, Juan Garassino began his professional journey in Buenos Aires by WORKING_AT a local architecture and urban design firm (ORGANIZATION) as a **BIM Design Coordinator (ROLE)** (TIME_PERIOD: 2018–2020). In this role, he was RESPONSIBLE_FOR implementing Building Information Modeling processes and coordinating multidisciplinary design teams. He managed digital architectural models and ensured that sustainable design principles were integrated into projects (ACHIEVEMENT), reflecting his commitment to environmentally conscious urban development.

During this period, Juan Garassino also embraced his identity as a "machine learning-based urbanist," engaging in early experiments that applied data science to city planning. For example, he **WORKED_ON** internal R&D projects where he USED generative algorithms to analyze urban traffic patterns and tested reinforcement learning methods like Proximal Policy Optimization (PPO) (SKILL) to simulate optimal city layouts. These innovative efforts demonstrated how artificial intelligence could improve urban efficiency and sustainability, foreshadowing modern smart city concepts and increasing his interest in AI-driven urban solutions. Notably, his ideas were inspired by global models like Singapore's Smart Nation initiative, where advanced technology and forward-thinking policies transformed the country into a global leader in smart infrastructure.

Transition to AI and Data Science

In 2021, driven to deepen his technical expertise, Juan Garassino relocated to Berlin (PLACE) – a dynamic European startup hub known for its vibrant tech scene and creative communities. He undertook intensive training to complement his architecture background with data science and machine learning skills. Juan Garassino STUDIED_AT **Le Wagon (ORGANIZATION)** in Berlin, completing its renowned Data Science & AI bootcamp (EDUCATION) in 2022. He EARNED a certification in data science, acquiring SKILLS in Python programming, statistical modeling, and deep learning. His capstone project at Le

Wagon, titled **“AI for Smart Cities” (PROJECT)**, reflected his passion for blending urban planning with AI. In this project, he applied machine learning to urban data – using neural networks and computer vision techniques to analyze city infrastructure and energy usage patterns – with the goal of informing more sustainable city policies. The approach echoed improvements seen in leading smart cities; for instance, AI-optimized public transport systems have reduced wait times and carbon emissions in places like Singapore. This hands-on project not only honed Juan’s technical abilities but also reinforced the real-world impact of AI on urban sustainability (ACHIEVEMENT).

Juan Garassino’s outstanding performance led to an invitation to join Le Wagon’s instructional team. He WORKED_AT **Le Wagon** as a **Deep Learning and MLOps Instructor (ROLE)** (TIME_PERIOD: Jan 2023 – Present), contributing to the bootcamp’s enterprise (B2B) training programs. In this role, he was RESPONSIBLE_FOR mentoring professionals and students in applied machine learning workflows across the entire ML lifecycle – from designing neural network models to deploying and monitoring them in production (HAS_SKILL in MLOps). He developed and delivered curriculum modules on deep learning and data engineering, sharing his dual-domain expertise in architecture and AI.

He also COLLABORATED_WITH Le Wagon’s Tokyo campus (PLACE) as a visiting instructor during early 2024 (TIME_PERIOD: Jan 2024 – Mar 2024), where he taught an immersive data science course to an international cohort. This experience showcased his adaptability and communication skills in a global context. Juan Garassino taught in English, the lingua franca of tech, and even utilized basic Japanese greetings with students, demonstrating cultural awareness. His success in this teaching role contributed to education in the tech community and highlighted his talent for conveying complex concepts across languages and cultures.

Generative AI Engineering and Leadership

In early 2024, Juan Garassino transitioned fully into the tech industry’s forefront by joining **Caidera (ORGANIZATION)**, a Berlin-based AI startup, as a **Founding Associate and Generative AI Engineer (ROLE)** (TIME_PERIOD: 2024 – Present). At Caidera, he WORKED_AT the cutting edge of applied artificial intelligence, where the company’s mission is to provide AI agent solutions for life sciences marketing teams. Juan contributes as a GenAI Engineer, driving innovation and enhancing AI products through his expertise. He is RESPONSIBLE_FOR designing and implementing large language model (LLM) systems and generative workflows that automate content creation while ensuring compliance with strict regulations in healthcare marketing. In this leadership capacity, Juan Garassino collaborates closely with the company’s co-founders (including Max Sieg and Daniel Fernau) and a multidisciplinary team of data scientists and domain experts, collectively building Caidera’s core product from the ground up.

As a founding associate, Juan Garassino is PART_OF strategic decision-making at Caidera and has helped shape the startup’s innovative culture. He also became integrated into Berlin’s startup ecosystem through the **Science & Startups (ORGANIZATION)** network, participating in events and mentorship programs that connect science-based entrepreneurs. During his tenure at Caidera, Juan has SPECIALIZED_IN advanced natural language processing and generative AI techniques for enterprise use. He fine-tunes transformer models for domain-specific tasks and implements robust MLOps pipelines, ensuring that the AI agents are reliable and secure. Under his technical guidance, Caidera’s AI platform has ACHIEVED significant improvements in clients’ marketing campaign efficiency, showcasing how generative AI can streamline real-world business processes. His role at Caidera illustrates a convergence of technical leadership and creative problem-solving, rooted in his diverse background.

Key Projects and Innovations

One of Juan Garassino's signature innovations is **PromptPlot (PROJECT)**, a creative research project he FOUNDED to explore the intersection of generative AI and physical art. PromptPlot transforms natural language prompts into physical drawings through a coordinated system of software and robotics. In this project, Juan Garassino WORKED_ON turning text into visuals by orchestrating a locally running large language model and a robotic pen plotter. He USED a pipeline wherein an LLM (powered by an engine like Ollama) streams real-time instructions to the plotter, effectively converting words into intricate, hand-drawn artwork. Under the hood, PromptPlot leverages a Python-based stack with **LlamaIndex (SKILL)** to manage and query language data efficiently, and it interprets user prompts to generate precise vector drawing commands. This technical foundation bridges natural language understanding and physical output, enabling the creation of tangible art without the need for extensive model fine-tuning. The PromptPlot journey builds on experiments Juan began four years prior with 3D Generative Adversarial Networks (GANs) for architectural space generation. Back then, tools were limited, but by combining his expanded toolkit in generative AI with his architectural imagination, he reimagined those early ideas in a new form. The result (RESULTED_IN) is a unique system that merges digital intelligence with artisan craftsmanship. Projects like PromptPlot demonstrate Juan Garassino's belief that AI's true potential lies beyond the obvious applications, venturing into uncharted territories that redefine creativity and manufacturing. This innovative project not only amplified human creativity by empowering designers and artists to turn ideas into reality, but also showcased Juan's ability to integrate art and engineering in practice.

Another notable endeavor is Juan Garassino's **"AI for Smart Cities" (PROJECT)** initiative, which highlights his ongoing commitment to sustainable urban innovation. In this self-driven research project, he CONTRIBUTED_TO the field of urban tech by prototyping AI solutions for smarter city management. Juan Garassino USED deep learning and reinforcement learning techniques (SKILL) to address real urban challenges: for instance, developing a model to optimize traffic signal timing and public transit scheduling based on real-time data, aiming to reduce congestion and energy consumption in city centers. He applied the PPO algorithm to train autonomous agents that simulate traffic coordination, and he utilized GANs to generate synthetic urban scenarios for stress-testing city infrastructure under various conditions. This project drew inspiration from global best practices; Juan studied how cities like Singapore leverage AI and data analytics to enhance urban life, noting tangible benefits such as improved public transport efficiency and reduced carbon emissions. By contextualizing similar AI strategies for the cities he knows (e.g. applying them to bus networks in Buenos Aires or traffic flows in Berlin), Juan Garassino's work ACHIEVED a blueprint for how AI can assist urban planners in making data-driven, sustainable decisions. The "AI for Smart Cities" project underscores his interdisciplinary approach – blending his urbanist insight with machine learning expertise – to create smarter, greener cities. It also exemplifies his drive to ensure that technological advancement results in social and environmental benefits, aligning with global sustainability goals.

Skills and Expertise

Juan Garassino's career is distinguished by a broad set of SKILLS that span design, technology, and education. He HAS_SKILL in traditional architectural design and **Building Information Modeling (BIM)**, grounded in his academic training and early career coordinating complex building projects. Complementing this, he possesses strong expertise in **Machine Learning (ML)** and **Deep Learning**. Juan is proficient in programming (especially Python) and has hands-on experience with ML frameworks like TensorFlow/Keras and PyTorch from developing various AI models. He SPECIALIZED_IN generative modeling techniques – including **Generative Adversarial Networks (GANs)** and diffusion models – which he employed for visual and spatial data projects, linking back to his design background. He is also

adept in **Natural Language Processing (NLP)** and LLM-based architectures; for example, at Caidera and in PromptPlot he integrated large language models into applications that interface with users and hardware. Additionally, Juan has developed considerable MLOps and data engineering skills, enabling him to deploy and maintain AI systems in production environments (a focus he also imparted as an instructor). He stays current with AI research and toolsets, experimenting with reinforcement learning algorithms (such as PPO for simulation and optimization tasks) and utilizing data visualization techniques to communicate insights. Beyond the technical realm, Juan Garassino is multilingual – he speaks Spanish (his mother tongue) and English fluently, has conversational proficiency in French from his time in Paris, and is learning German while living in Berlin. These language skills enhance his collaboration in international teams and reflect his adaptability in diverse environments.

Collaborations and Contributions

Collaboration has been a cornerstone of Juan Garassino's journey, as he frequently works across disciplines and cultures to achieve innovative outcomes. He has COLLABORATED_WITH professionals from varied fields – architects, software engineers, data scientists, and business strategists – to bring complex projects to fruition. Notably, Juan Garassino collaborated with his brother (PERSON), an electronic musician and sound designer based in Berlin, on a cross-disciplinary creative project. Together, the Garassino brothers developed an interactive installation that fused generative visual art with music: Juan's algorithms would produce real-time visuals in sync with music composed by his brother. This collaboration blended their respective expertise in technology and art, and it underscores Juan's personal commitment to exploring creativity without boundaries. In the open-source community, Juan Garassino has also CONTRIBUTED_TO projects on GitHub, sharing code and ideas. For example, he published a repository called "deepSculpt" (PROJECT) demonstrating 3D generative techniques for sculpture, and he has contributed improvements to AI toolkits, reflecting his ethos of knowledge sharing. He has additionally AUTHORED technical content to disseminate his insights. A notable example is his LinkedIn article *"Beyond Chatbots: When GenAI Engineering Meets Artistic Expression,"* in which he detailed the vision and technical underpinnings of the PromptPlot project. Through such publications (PUBLICATION), Juan engages with the broader tech and design communities and establishes himself as a thought leader at the intersection of AI and design.

In his educational and mentoring roles, Juan Garassino has directly contributed to building others' capacities. As an instructor at Le Wagon, he guided dozens of aspiring data scientists and engineers, many of whom have since launched their own careers – an indirect but meaningful ACHIEVEMENT of his influence on the next generation of technologists. He continues to participate in community knowledge-sharing, speaking at tech meetups, hackathons, and design workshops about topics like generative AI, smart cities, and ethical AI use. Juan's collaborative approach is also evident in the workplace, where he often took on informal leadership and coaching duties. He has MANAGED student project teams and coordinated with colleagues on interdisciplinary projects, always aiming to foster an environment of learning and innovation. His leadership style emphasizes empathy, creativity, and pragmatic problem-solving – traits likely honed from working in diverse cultural settings in Argentina, France, Germany, and Japan. Through these collaborations and contributions, Juan Garassino has built a reputation as a connector of ideas and people, dedicated to advancing technology in a way that is inclusive and impactful.

Ongoing Endeavors and Vision

As of 2025, Juan Garassino continues to push the frontiers of interdisciplinary innovation. DURING his current tenure at Caidera, he remains focused on advancing generative AI applications that have positive real-world impact, particularly in highly regulated and meaningful domains like healthcare

communication. Simultaneously, he is exploring new domains such as AI safety and red-team cybersecurity for AI systems – indicating his commitment to a holistic understanding of the technologies he develops. This recent interest in AI security involves testing and fortifying AI models against biases and vulnerabilities (SKILL), which complements his existing expertise by ensuring the robustness of AI solutions. Juan Garassino's journey – from designing buildings in Buenos Aires to engineering intelligent systems in Berlin – exemplifies the power of lifelong learning and the synergy between art and science. He has ACHIEVED a unique professional profile that bridges creative design and cutting-edge technology, and he envisions a future where these fields converge even further. By integrating architectural imagination with artificial intelligence, Juan Garassino aims to contribute to more sustainable, intelligent, and human-centric environments. His story is one of continuous growth and collaboration, rich with entities and relations: an architect-turned-AI-engineer who has STUDIED_AT top institutions, WORKED_AT innovative organizations, HAD diverse ROLES, and **WORKED_ON** transformative PROJECTS. This comprehensive blend of experiences positions Juan Garassino as a forward-thinking leader, committed to leveraging technology for creative and societal advancement.